

SKB HAIR DOCTOR: A Premium Barber Shop & Salon Management Ecosystem

Project Overview

SKB HAIR DOCTOR is a premium, high-performance, and fully responsive mobile marketplace solution engineered to bridge the gap between traditional grooming services and modern digital appointment commerce. Built using the **Flutter framework** and a reactive **Firebase backend**, the application provides a robust, scalable infrastructure designed to deliver a seamless user experience. The platform replaces manual bookings with an automated scheduling environment, dynamic service exploration, and real-time operational state tracking.

Technical Stack & Architecture

- **Frontend Framework:** Flutter & Dart (Cross-Platform iOS & Android compilation)
- **Backend Cloud Infrastructure:** Cloud Firestore (Real-time NoSQL Data Synchronization)
- **Identity & Access Management:** Firebase Authentication (Secure User Session Lifecycle)
- **Local State Caching:** Shared Preferences (Persistent session tokens and fast boot handling)
- **UI/UX Design Paradigm:** Advanced Glassmorphic Layout utilizing Custom Layer Blurring (BackdropFilter)

Core System Features & Modules

1. Real-Time Operational State Engine (Shop Status)

- **Reactive Synchronization:** Connected directly via an asynchronous pipeline to a single-document administrative state in Cloud Firestore.
- **Instant UI Feedback:** Whenever the administrator toggles the business operational state from the admin dashboard, the client application immediately renders the change without requiring a page refresh or manual pull-to-refresh action, flashing a high-visibility **"SHOP OPEN" (Green Accent)** or **"SHOP CLOSED" (Red Accent)** indicator.

2. Intelligent Dynamic Services Marketplace Grid

- **Admin-Driven Content Assets:** The home dashboard features a dynamically rendered grid layout. When a new service is published via the admin panel, the client application dynamically fetches the dynamic pricing (PKR) and remote image assets via the Cloud Storage reference URL (`imageUrl`).

- **Fault-Tolerant Fallback System:** To enforce bulletproof structural integrity and prevent app crashes under poor network conditions, a custom fallback mechanic is implemented. If no remote image is present or fails to resolve, the card UI gracefully switches to preset contextual iconography (brush, spa, cut, face).
- **Responsive Multi-Axis Scaling:** Engineered with safe aspect ratio boundaries to guarantee that text strings, currency numbers, and graphical assets scale seamlessly across varying screen form-factors, eliminating layout constraint overflows.

3. Integrated Micro-Interaction Booking Engine

- **Contextual Bottom Sheet:** Users can initiate an appointment request instantly via a high-performance, interactive bottom sheet container without disrupting their current navigation stack.
- **Asynchronous Execution:** Built with asynchronous Dart processing, ensuring that the scheduling pipeline executes in the background with zero blocking overhead on the main UI threads.

4. Live Profile & Modular Navigation Drawer

- **Real-Time Identity Injection:** Utilizes a persistent StreamBuilder pipe connected to the active authenticated user document, displaying live updates of the user's name, email, and security profile avatar.
- **High-Performance Routing:** Provides localized entry points to distinct app modules including:
 - *Profile Personalization Subsystem*
 - *Centralized Appointment Tracking Dashboard*
 - *Exclusive Hair Care E-Commerce Marketplace*
 - *Integrated Geolocation Shop Tracker*
 - *Stylist Portfolio & Trend Gallery*
 - *Privacy Compliance & Terms Document Documentation*

Security & Session Management

- **Session Token Sanitization:** Integrated with an explicit `SharedPreferences.clear()` sweep mechanism triggered upon user logout.
- **State Protection:** This ensures all cached session credentials, identity metrics, and security indices are instantly flushed from the local hardware memory space, preventing unauthorized application state hijacking.

ADMIN PANEL



WE TREAT YOUR HAIR

WELCOME BACK

LOGIN

@ Email Address

🔒 Password

[Forgot Password?](#)

LOGIN

Don't have an account? [Sign Up](#)



WE TREAT YOUR HAIR

SIGN UP

👤 Full Name

@ Email Address

📞 Phone Number

🔒 Password

CREATE ACCOUNT

Already have an account? [Sign In](#)

☰

SKB ADMIN PANEL

BUSINESS SUMMARY



PKR 4650

REVENUE



1

PENDING



2

CUSTOMERS



3

SERVICES

RECENT BOOKINGS

View All >

M

Muhammad Abdullah

sfg (x1)

PENDING

M

Muhammad Abdullah

sfg (x1)

ACCEPTED

B

Bilawal Rehman

sfg (x1), Hair Cut (x1)

ACCEPTED

M

Muhammad Abdullah

sfg (x1), Hair Cut (x1)

ACCEPTED

M

Muhammad Abdullah

sfg (x1)

ACCEPTED

☰

SKB ADMIN PANEL

Payment Receipt

×

4:47

34%

←



WE TREAT YOUR HAIR

SIGN UP



Full Name



Email Address



Phone Number



Password

CREATE ACCOUNT

Already have an account? [Sign In](#)



SKB ADMIN PANEL

PENDING REQUESTS

FETCHING NAME...

07:30 PM

Services: sfg (x1)

Barber: Abdullah Time: 30 mins

Price: PKR 150



VIEW PAYMENT RECEIPT

APPROVE

REJECT



SKB ADMIN PANEL

PENDING REQUESTS

FETCHING NAME...

07:30 PM

Services: sfg (x1)

Barber: Abdullah Time: 30 mins

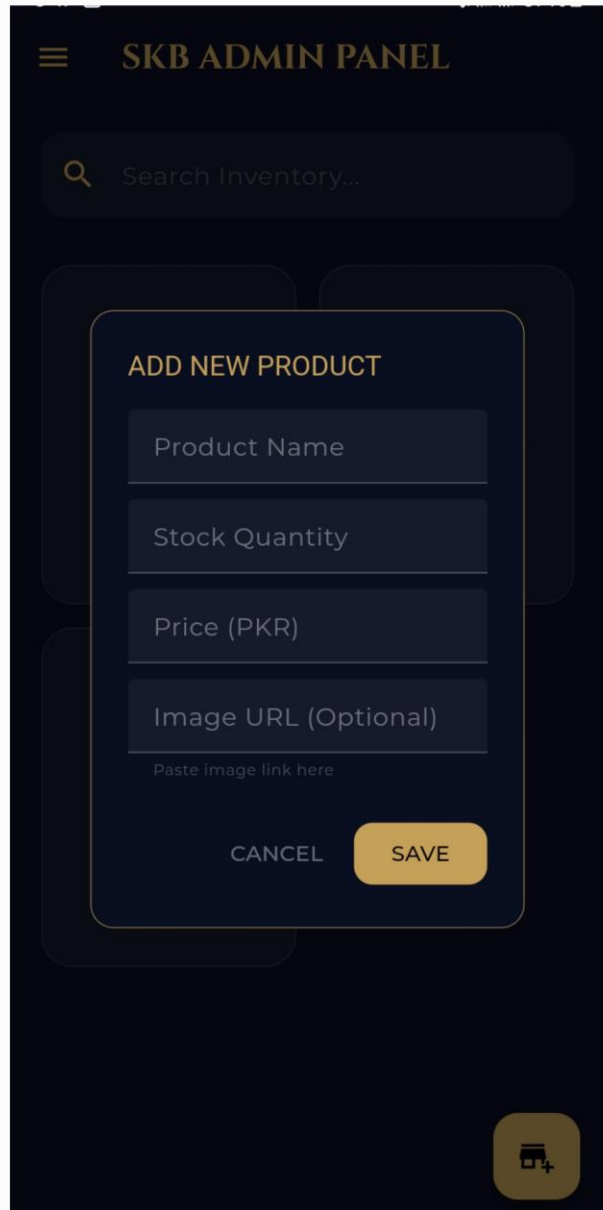
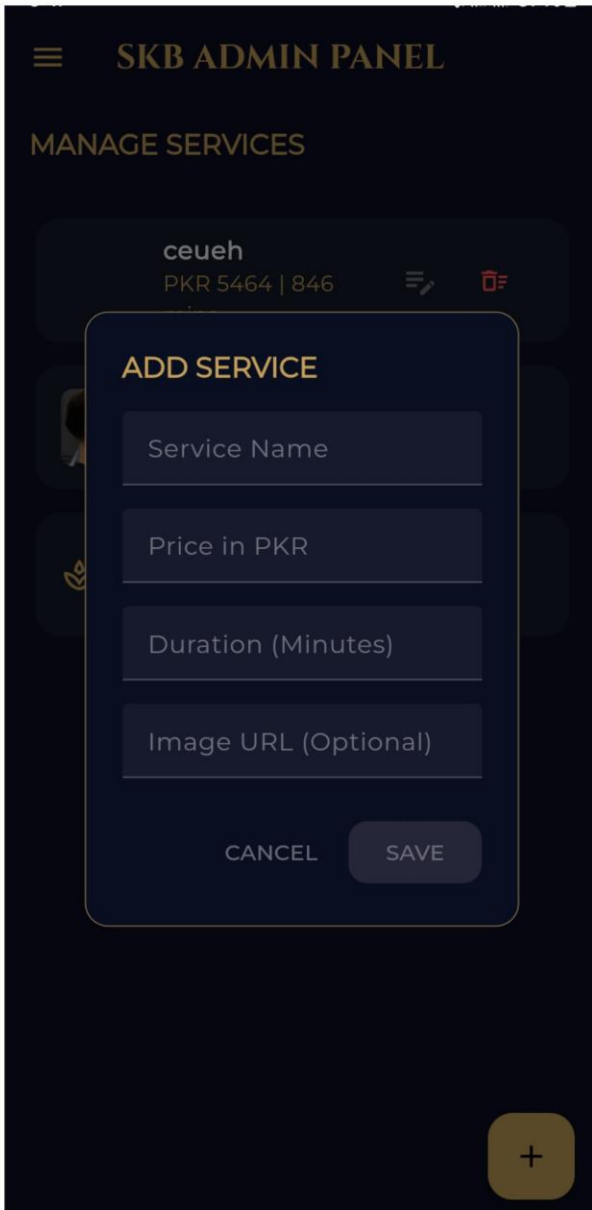
APPOINTMENT CANCELLATION

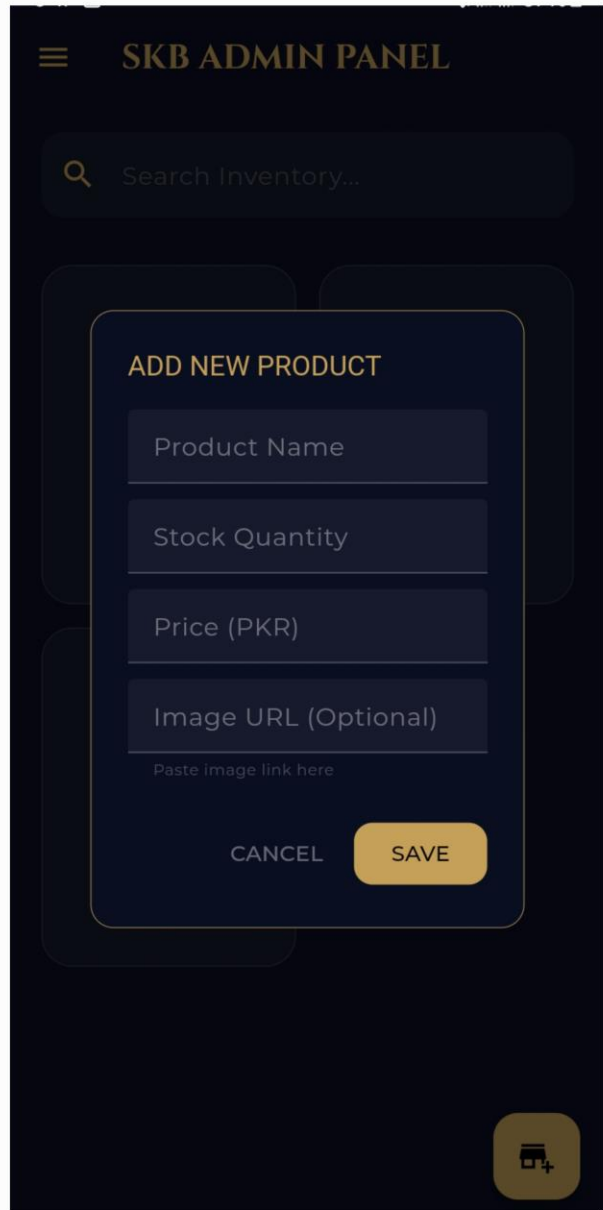
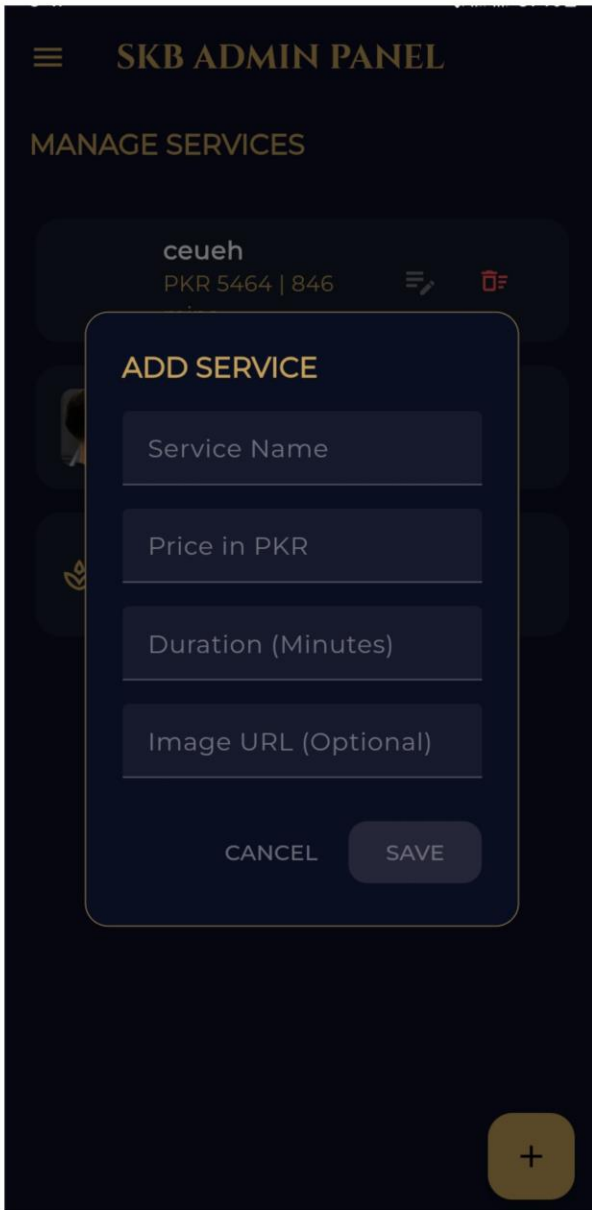
State the reason why you are rejecting this appointment:

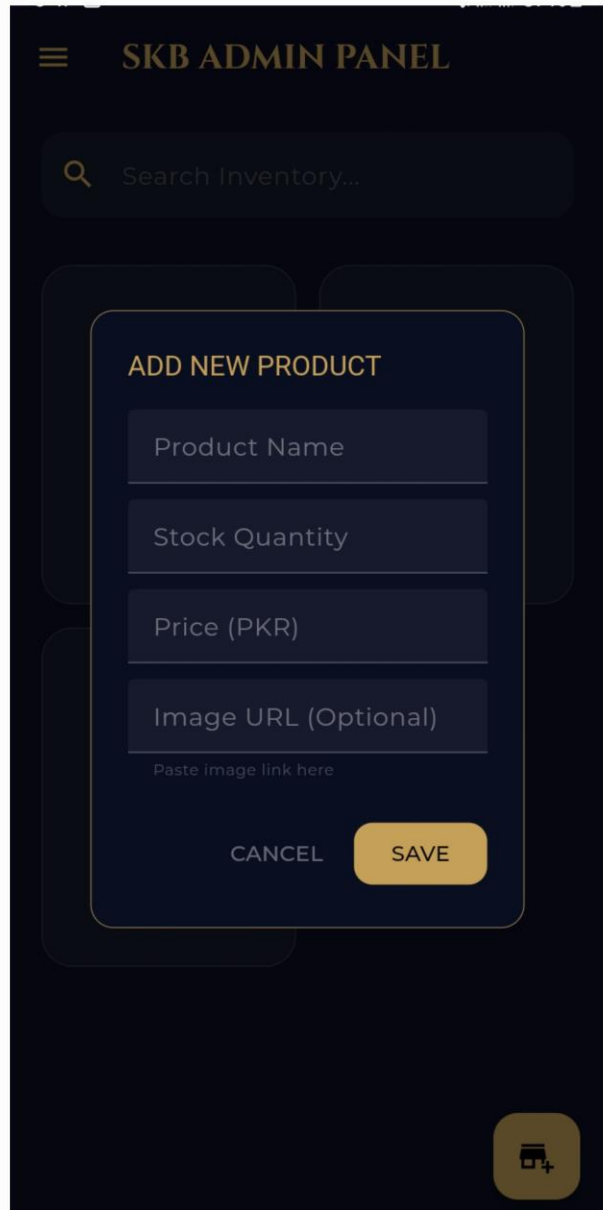
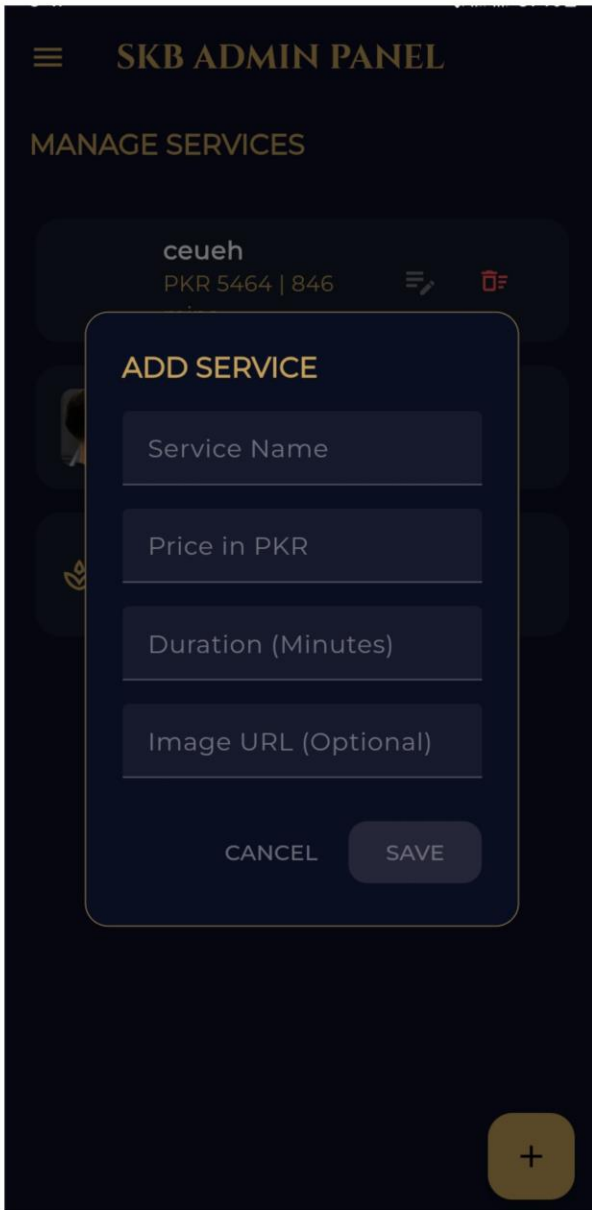
e.g., Slot time clash or Barber unavailable...

BACK

CANCEL BOOKING







←

SHOP DETAILS & TIMINGS

SHOP OPERATIONAL HOURS

Opening Time

8:00 AM

🕒

Closing Time

8:00 PM

🕒

GENERAL INFORMATION

📍 ADDRESS

Canal Road Near Family Hospital

🗺 MAPS URL

https://maps.app.goo.gl/fJn8U9e'

📞 WHATSAPP NUMBER

📞 SUPPORT NUMBER

SAVE CONFIGURATIONS

<

STAFF MANAGEMENT



Abdullah

Exp: 10 Years

☒

🗑



Bilawal Rehman

Exp: 10 Years

☒

🗑



Khawar Rehman

Exp: 10 Years

☒

🗑

+

USER PANEL

Testing Email: muhabdullah492@gmail.com

Password: Ali1231@



WE TREAT YOUR HAIR

WELCOME BACK

LOGIN

@ Email Address

🔒 Password

[Forgot Password?](#)

LOGIN

Don't have an account? [Sign Up](#)



WE TREAT YOUR HAIR

SIGN UP

👤 Full Name

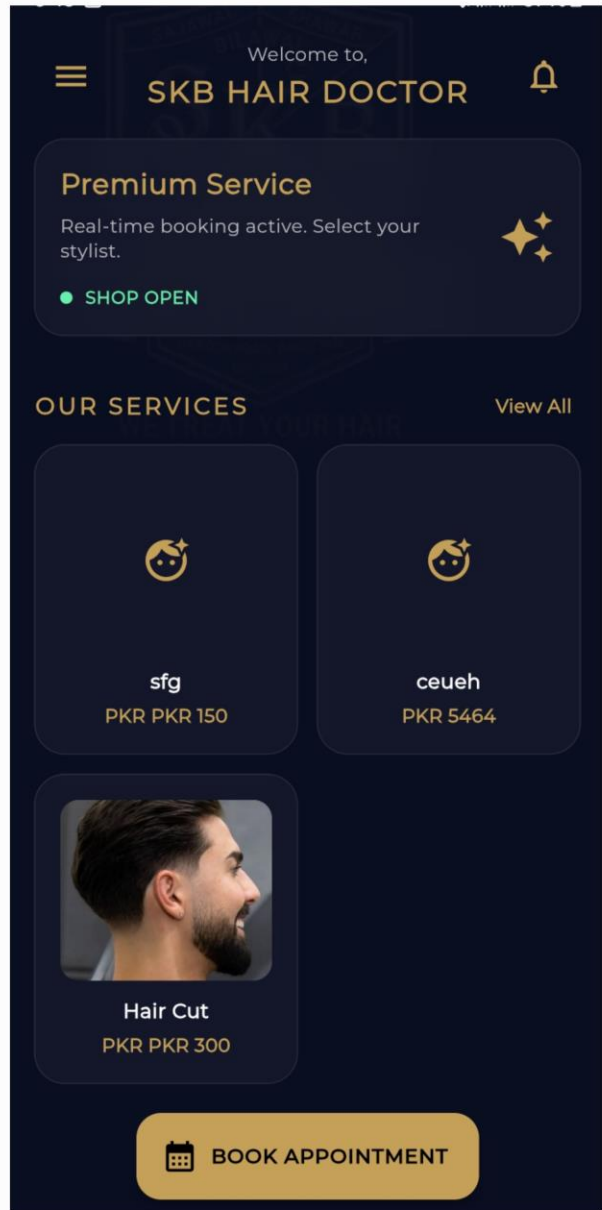
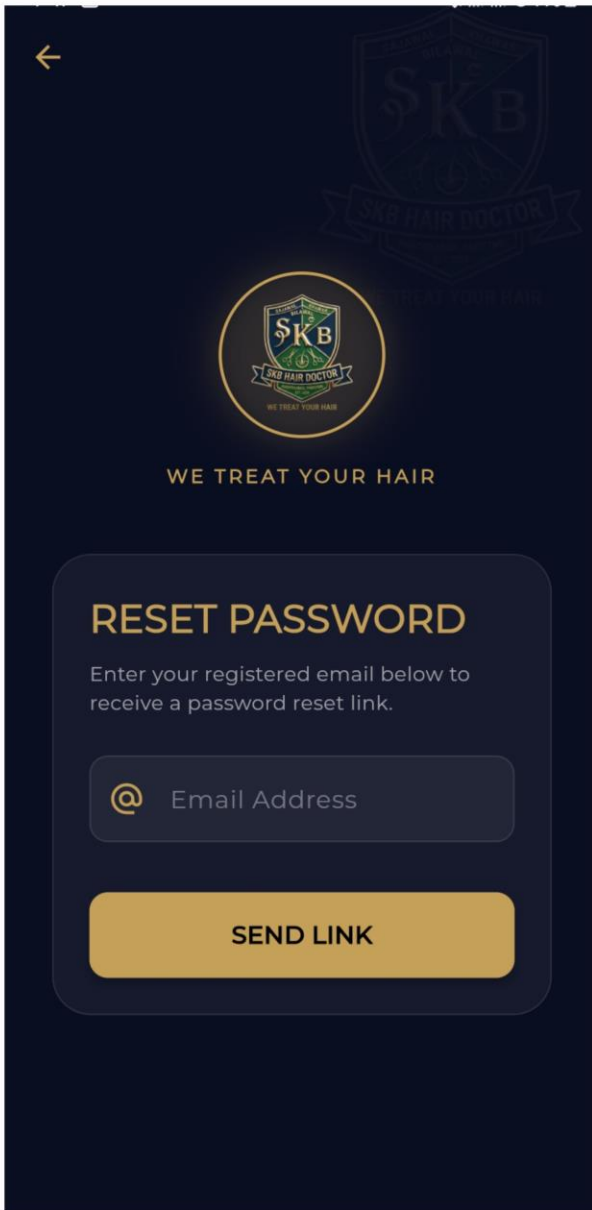
@ Email Address

☎ Phone Number

🔒 Password

CREATE ACCOUNT

Already have an account? [Sign In](#)



BOOK YOUR SLOT

1. Select Date

Monday, 22 June 2026



2. Select Barber



Bilawal Rehm...
10 yrs



Khawar Reh...
10 yrs



Abdullah
10 yrs

3. Select Services (Quantity)

sfg

PKR PKR 150 | 30 mins

— 1 +

ceueh

PKR 5464 | 846 mins

— 0 +

Hair Cut

PKR PKR 300 | 60 mins

— 0 +

4. Available Time Slots

06:30 PM

07:00 PM

07:30 PM

08:00 PM

3. Select Services (Quantity)

sfg

PKR PKR 150 | 30 mins

— 1 +

ceueh

PKR 5464 | 846 mins

— 0 +

Hair Cut

PKR PKR 300 | 60 mins

— 0 +

4. Available Time Slots

06:30 PM

07:00 PM

07:30 PM

08:00 PM

5. Summary

Total Estimated Time

30 mins

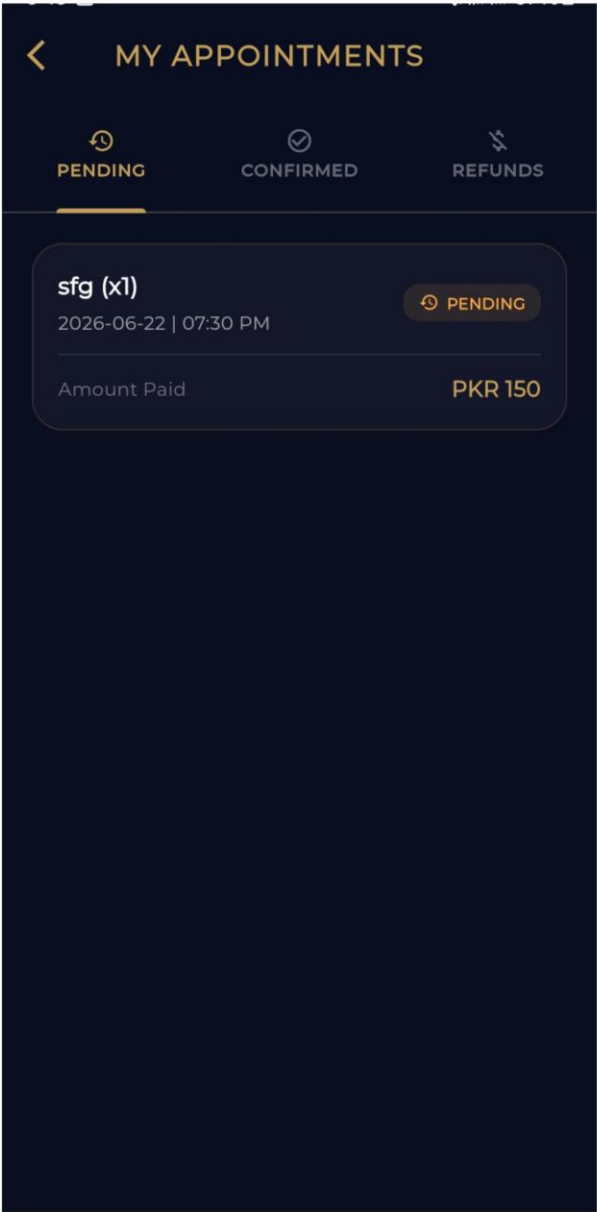
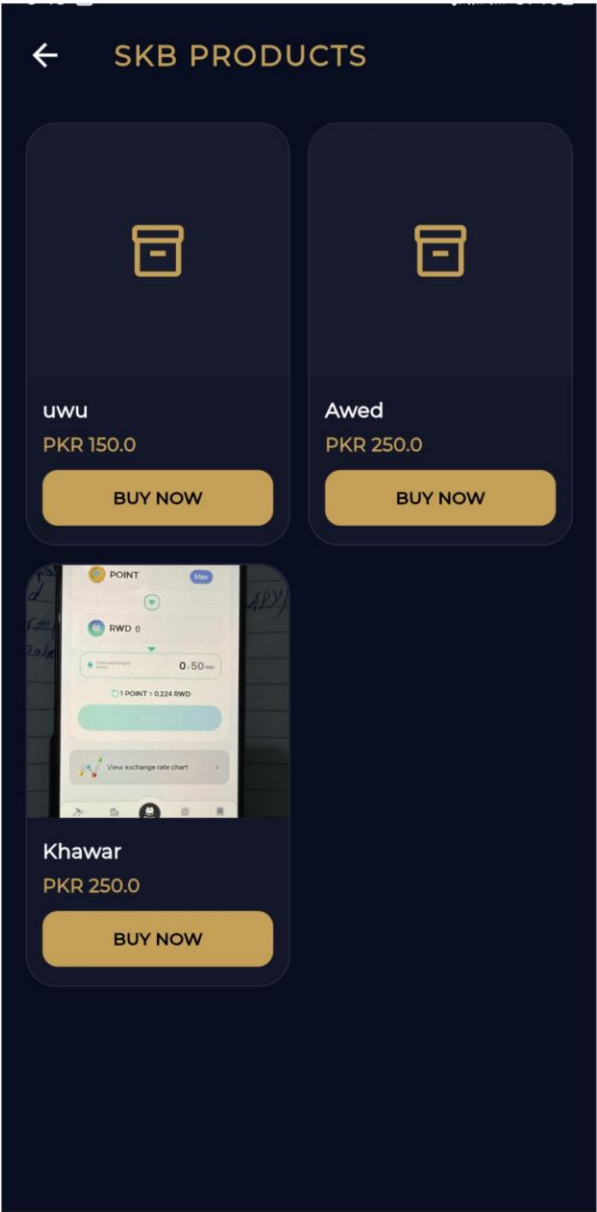
Total Price

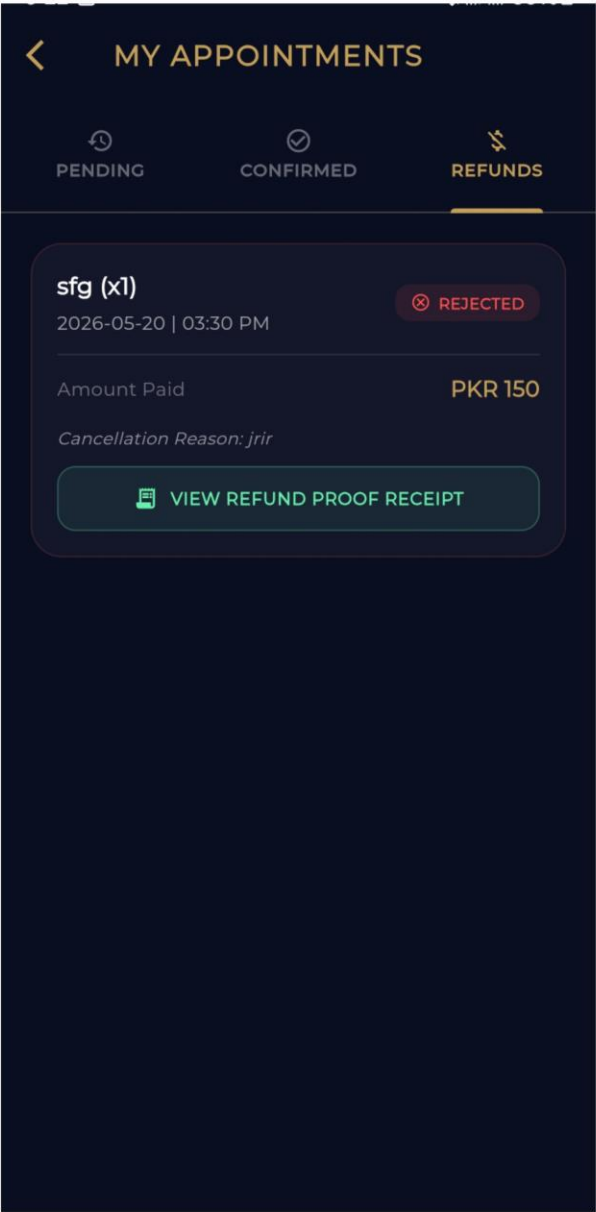
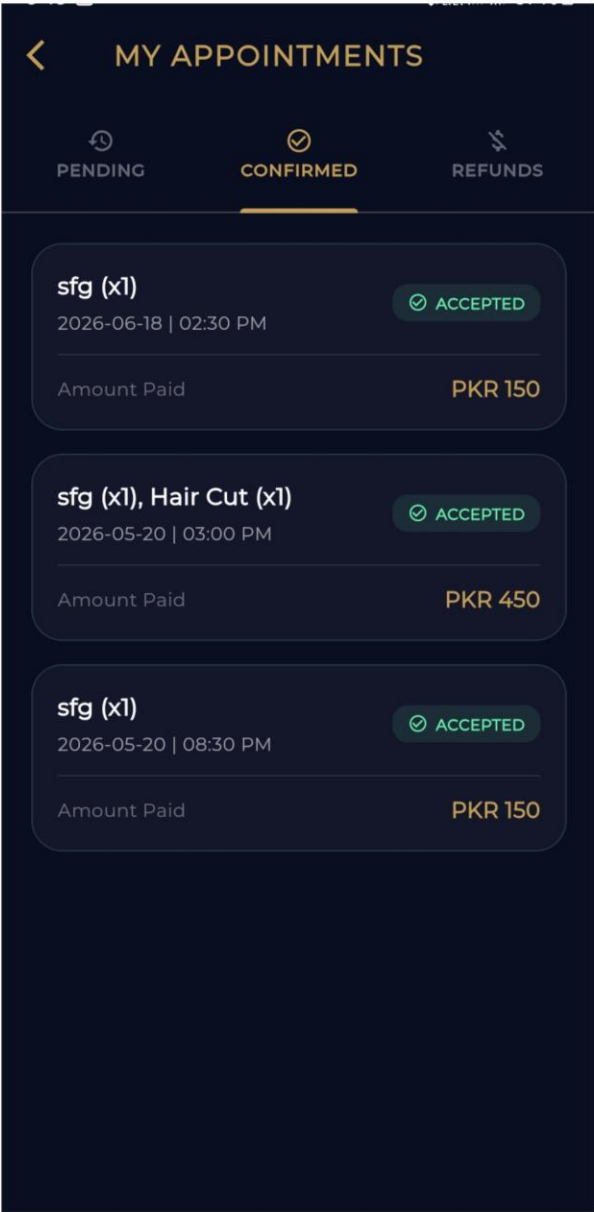
PKR 150

Advance Payment Req (20%)

PKR 30

PROCEED TO PAYMENT







SHOP LOCATION



Google Map View

SKB Luxury Salon

📍 Main Bazar Road, Near Sarwat, Pakistan

🕒 Open Daily: 10:00 AM - 10:00 PM

Quick Actions

▲ DIRECTIONS

💬 WHATSAPP



OUR GALLERY

